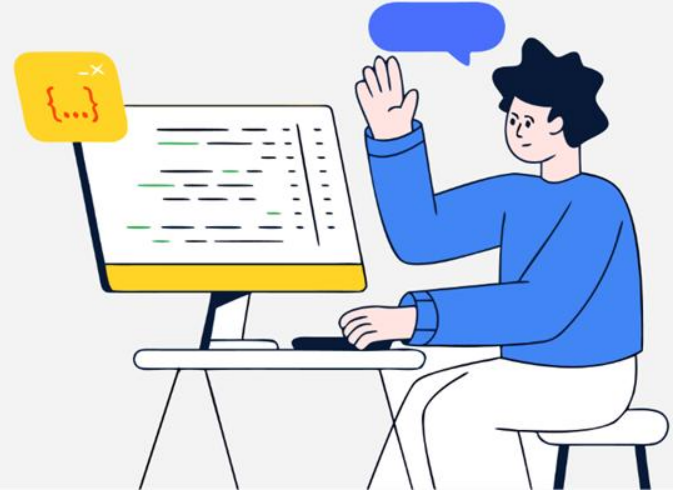




Solution Challenge



Team Details

- a. Team name: Algo Masters
- b. Team leader name: Vishal Dhakad
- c. Team members: Radhika Gupta, Mahima suryavanshi, Ayushi Dhote
- d. Problem Statement : Data Driven Volunteer Coordination for Social Impact
- e. Theme :Smart Resource Allocation [Open Innovation]

NEXUS: AI-Powered Intelligent Response Ecosystem

Core Concept A unified platform that transforms fragmented community reports into structured, real-time action. NEXUS replaces slow, manual processes with an automated system that identifies, prioritizes, and coordinates disaster and social relief efforts.

Key Mechanisms:

- **Intelligence:** Uses Google Gemini to analyze multi-modal inputs (voice, text, images) and automatically detect urgency levels.
- **Visualization:** A Live Urgency Intelligence Map provides decision-makers with a color-coded heatmap of critical needs.
- **Coordination:** An AI Volunteer Engine matches the nearest skilled responders to verified tasks instantly.

Impact & Accessibility:

- **Multi-Tenant Design:** Provides secure, separate workspaces for NGOs, government bodies, and researchers on a single infrastructure.
- **Accountability:** Ensures transparent resolution through mandatory proof-of-work uploads and AI-driven duplicate detection.

1. Differentiation: Why NEXUS?

- **Proactive Intelligence:** Uses AI heatmaps to predict trends rather than just recording past issues.
- **Unified Ecosystem:** Connects citizens, AI, NGOs, and Government on one collaborative platform.
- **Actionable Insights:** Automatically prioritizes urgency and filters duplicates to eliminate data noise.

2. Impact: How it Solves the Problem

- **Structured AI Data:** Converts voice, images, and text into a categorized database via Google Gemini.
- **Geospatial Visibility:** Maps every report onto a live dashboard for instant situational awareness.
- **Rapid Coordination:** An automated engine matches volunteers to tasks by skill and location instantly.
- **Closed-Loop Transparency:** Ensures accountability through mandatory proof-of-completion uploads.

3. Unique Selling Propositions (USPs)

- **AI Urgency Detection:** Automated High/Medium/Low priority classification.
- **Multi-Tenant Architecture:** Secure, private workspaces for multiple NGOs on a single platform.
- **Inclusive Reporting:** Supports multi-modal input (Voice/Images) to overcome literacy barriers.
- **Live KPI Dashboard:** Real-time monitoring of active cases, resolution times, and impact.

Solution

AI & Smart Logic

- **Automated Categorization:** Uses Google Gemini to analyze and label community reports.
- **Urgency Detection:** Automatically ranks issues (High, Medium, Low) for prioritized response.
- **Duplicate Filtering:** AI-driven detection of repeated reports to prevent redundant resource usage.

Live Operations & Mapping

- **Interactive Crisis Map:** Real-time geospatial visualization of hotspots with color-coded markers.
- **Live Admin Dashboard:** Unified view for tracking resolution rates, active cases, and impact metrics.
- **Volunteer Matching:** Automated engine to assign tasks based on proximity and specific skill sets.

Field Execution & Verification

- **Multi-Modal Input:** Support for Text, Voice notes, and Image-based reporting to ensure inclusivity.
- **Task Status Tracking:** End-to-end lifecycle management (Assigned → In-Progress → Completed).
- **Proof of Completion:** Mandatory image/document uploads for resolution verification and transparency.

Enterprise Architecture

- **Multi-Tenant Workspaces:** Isolated and secure data environments for different organizations on one platform.
- **Role-Based Permissions:** Granular access control for Admins, Field Volunteers, and Researchers.

Process Flow Diagram / Use-Case Diagram

User / Citizen / NGO Staff

↓
Submit Complaint (Text / Voice / Image / Form)

↓
AI Analysis Engine
(Category + Urgency Detection)

↓
Store Data in Firebase Database

↓
Live Dashboard Update (Map + Reports + Analytics)

↓
Volunteer Matching System (Nearest / Skilled Volunteer)

↓
Task Assigned

↓
Volunteer Accepts Task

↓
Field Work Completed

↓
Upload Proof (Image / Notes)

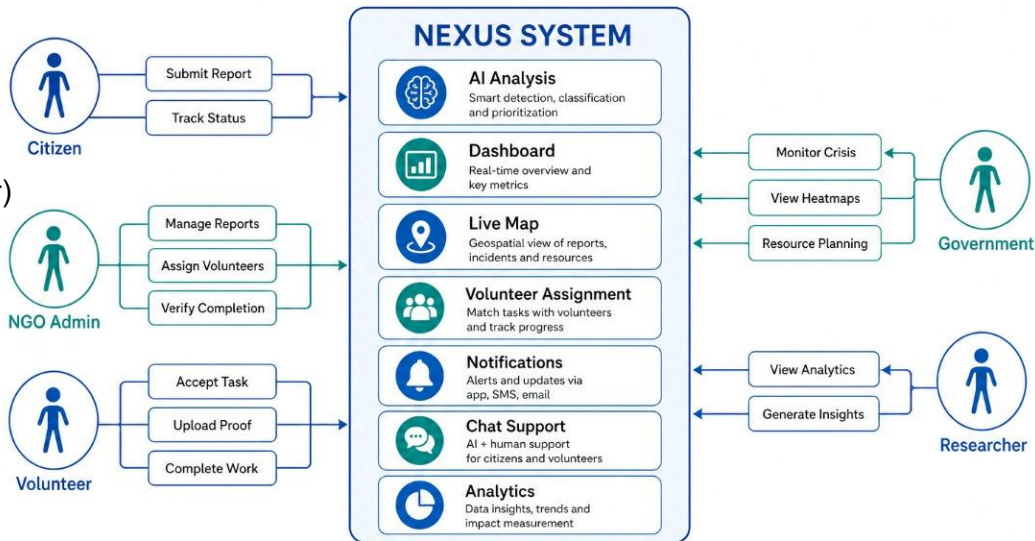
↓
Admin Verification

↓
Issue Marked Resolved

↓
Reports & Insights Generated

NEXUS Civic Intelligence Platform

USE CASE DIAGRAM



NEXUS

Stronger Communities. Smarter Decisions.



Architecture diagram



Frontend & User Interface

- **React.js:** For building a dynamic, responsive, and component-based web interface.
- **Tailwind CSS:** Modern utility-first CSS framework for rapid and clean UI styling.
- **React Router:** Managing seamless navigation between the dashboard and reporting pages.

Backend & Cloud Infrastructure

- **Firebase Firestore:** Real-time NoSQL database for instant data synchronization across clients.
- **Firebase Authentication:** Secure user management for NGOs, volunteers, and admins.
- **Firebase Storage:** Scalable cloud storage for media reports and proof-of-completion files.
- **Firebase Hosting:** Global CDN deployment for high availability.

Artificial Intelligence (Google Ecosystem)

- **Google Gemini API:** Core intelligence for multi-modal analysis (text/image/voice) and automated urgency classification.
- **NLP Engines:** Processing unstructured community feedback into categorized, actionable data points.

Geospatial & Mapping

- **React Leaflet:** For rendering interactive, high-performance maps.
- **Geolocation API:** Capture precise coordinates for community report accuracy.

Analytics & Visualization

- **Recharts / Chart.js:** Transforming raw database metrics into visual heatmaps and performance charts.
- **KPI Tracking:** Custom dashboards for monitoring resolution times and regional impact.

Phase 1: Resilience & Accessibility

- **Offline-First Architecture:** Transitioning to a local-first sync engine to enable field reporting in zero-connectivity zones with automatic background synchronization.
- **Multilingual Support (Bhashini/Sarvam AI):** Integrating voice-to-text for 11+ regional Indian languages to bridge literacy and linguistic gaps.
- **Omnichannel Reporting:** Deploying lightweight WhatsApp, Telegram, and SMS bots to allow citizens to report issues without needing the full web app.

Phase 2: Advanced Intelligence & IoT

- **Predictive Crisis Modeling:** Moving from reactive to proactive by using historical data to forecast seasonal risks (e.g., recurring water shortages or flood-prone areas).
- **Vision-Language Models (VLM):** Enhancing AI to automatically assess the scale of physical damage from user-uploaded photos for better resource estimation.
- **IoT Sensor Integration:** Connecting smart water-level and weather sensors to trigger “High Urgency” alerts automatically on the NEXUS dashboard.

Phase 3: Trust, Security & Scaling

- **Blockchain Audit Trail:** Implementing a decentralized ledger for “Proof of Completion” to ensure transparent, tamper-proof reporting for donors and stakeholders.
- **Aadhaar/Digital ID Integration:** Securely onboarding and verifying NGO staff and volunteers through government-authorized identity APIs.

Provide links to your:

1. GitHub Public Repository : <https://github.com/Radhika-Gupta894/nexus-platform>
2. Demo Video Link :-
https://drive.google.com/file/d/1kP0iPywJvKk1ysMIJ_S2Rb_OMKJy2hgw/view?usp=drivesdk
3. MVP Link:- <https://agent-69ed9fc99d61577fb9--lovely-vacherin-1014d9.netlify.app>

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Solution Challenge



Thank You